

29

B

$^{28}_{8}\text{O}$: 8 protons, 20 neutrons (doubly magic)

$^{40}_{20}\text{Ca}$: 20 protons, 20 neutrons (doubly magic)

$^{56}_{26}\text{Fe}$: 26 protons, 30 neutrons (**not** doubly magic)

$^{50}_{28}\text{Ni}$: 28 protons, 22 neutrons (**not** doubly magic)

$^{126}_{50}\text{Sn}$: 50 protons, 76 neutrons (**not** doubly magic)

30

A

From the figure, background count rate can be estimated to be 8 s^{-1}